

David Mailland¹, Petro Kosobutskyy²

¹ Independent Researcher, formerly Reliability System Engineer, Thales Alenia Space, France,
E-mail: david.mailland@protonmail.com, ORCID 0009-0002-8480-1325

² Computer Design Systems Department, Lviv Polytechnic National University,
12, S. Bandery str., Lviv, Ukraine, E-mail: petkosob@gmail.com, ORCID 0000-0003-4319-7395

MODELLING THE COLLATZ PROBLEM FROM A JACOBSTHAL VIEWPOINT

Received: February 17, 2026 / Accepted: April 01, 2026 / Published: April 30, 2026

© Mailland D., Kosobutskyy P., 2026

<https://doi.org/10.23939/cds2026.01.049>

Abstract. This work does not aim to solve the Collatz conjecture, but to reformulate it within a new analytical framework based on Jacobsthal sequences. This reformulation reveals structural regularities and symmetries that remain hidden in the traditional approach, offering a new mathematical perspective from which the Collatz-type problems may be further explored. This approach can be reproduced in System Engineering analyses, where alternative representation are often valuable for exposing hidden structure and improving understanding. Beyond its mathematical significance, the proposed framework may also offer conceptual and methodological benefits for the field of Information Technology (IT). Many problems in computer science involve the analysis of iterative processes, recursive algorithms, and discrete state transitions. The Collatz transformation itself can be viewed as a simple computational process governed by conditional rules and repeated iteration, which makes it an interesting model for studying algorithmic behavior. Sequence-based representations are widely used in areas such as data compression, cryptography, and pseudo-random number generation. The identification of structured recurrence relations and parity patterns within integer sequences can provide insights into the design of efficient algorithms for generating deterministic yet complex numerical sequences. Such sequences are frequently employed in cryptographic systems, simulation models, and distributed computing applications.

Keywords: model-based analysis, discrete dynamical systems, Collatz conjecture, Jacobsthal sequences, inverse problem formulation, structural regularities, modular arithmetic, 2-adic valuation, reproducible computation.

Introduction

Sequence-generating algorithms are frequently used, both with deterministic inputs and with stochastic components when modelling complex systems using computer and information technologies. A classical example of a deterministic rule that produces unexpectedly complex behaviour is the Collatz iteration, which can be viewed as a simple sequence generator defined on the positive integers.

The Collatz problem (also known as the $3n+1$ problem) is one of the most famous open questions in Mathematics. The classical conjecture claims that, starting from any positive integer, repeated iteration eventually falls into the trivial cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ [1]. Therefore, it is of considerable interest for research.

Review of Modern Information Sources on the Subject of the Paper

The majority of approaches are implemented in a descending way: one starts from a given integer and follows its trajectory step by step until it reaches a small value. The Collatz iteration defines a deterministic, memoryless update rule: each next value depends only on the current integer (i.e., it satisfies the Markov property), although it is not a stochastic Markov chain unless transition probabilities are introduced [2–3].

Although the Collatz problem has been investigated in a wide range of ways [4–7], recent studies have also explored Collatz-type dynamics through inverse and graph-based formulations [8–10]. In a recent modelling study based on Petri nets [11], the Collatz problem was approached from an inverse perspective. This viewpoint is particularly appealing because it allows inverse relations to be organised into explicit tree-like structures, making branching mechanisms and global organisation more apparent.

A complementary viewpoint was developed in [8–10], where the set of natural numbers is first organised into families of sequences induced by the binary decomposition. Within these sequences, one can identify a periodic set of special points (nodes) related to Jacobsthal-type numbers. These nodes generate a branching tree structure; read in the reverse direction, the branches naturally recover Collatz trajectories.

Objectives and Problems of Research

In this article, we adopt a different viewpoint: instead of following trajectories downward, we construct a global structure in the opposite direction.

This construction is based on simple arithmetic expressions related to Jacobsthal-type sequences. It produces a branching structure, which we call a Jacobsthal tree. Remarkably, when this tree is read in reverse, its branches recover trajectories of Collatz type.

The purpose of this paper is not to introduce a new theory nor to solve the Collatz conjecture. Our goal is to present, at an elementary and intuitive level, the core ideas of this Jacobsthal-based approach. We focus on the classical case $\kappa = 3$, corresponding to the usual $3n+1$ rule, and emphasize concrete examples, visual intuition, and simple arithmetic arguments.

Main Material Presentation

Following standard notation, we denote by $C: \mathbb{N} \rightarrow \mathbb{N}$ the classical Collatz map:

$$C(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod{2} \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2} \end{cases} \quad (1)$$

In addition, in order to remove the even branches and work with a trajectory consisting only of odd integers, we introduce the accelerated (odd-only) map:

$$T: \mathbb{N}_{\text{odd}} \rightarrow \mathbb{N}_{\text{odd}} \quad T(n) = \frac{3n + 1}{2^{v_2(3n + 1)}}, \quad (2)$$

where $v_2(x)$ denotes the 2-adic valuation of x (i.e., the largest integer $k \geq 0$ such that 2^k divides x). Equivalently, writing $k = v_2(3n + 1)$, one obtains the Terras (accelerated Collatz) function [2]:

$$T(n) = \frac{3n + 1}{2^k}, \quad k = v_2(3n + 1). \quad (3)$$

Function (3) removes even iterates from (1) by applying one $3n + 1$ step and then dividing by 2 until an odd integer is obtained. Hence, each application of T corresponds to one $3n + 1$ step followed by all possible halving steps.

The Collatz conjecture states that, for every initial number $n_0 \in \mathbb{N}$, the iterates of C eventually reach the cycle: $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. This statement is known as the Collatz conjecture, which, despite intensive research, has not yet been proved. A short computation illustrates the mechanism. Starting from $n_0 = 6$, we obtain: $6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

Starting from $n_0 = 7$, we obtain:

$$7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow \dots \rightarrow 1$$

Even for small initial values, the iteration may rise before eventually decreasing. This combination of simplicity (a two-branch rule) and irregular-looking behaviour is precisely what makes the problem so challenging.

A key ingredient in the Jacobsthal-based viewpoint is that every positive integer can be written in a unique way as an odd number times a power of two. More precisely, for every $n \in \mathbb{N}$ there exist a unique odd integer θ and a unique exponent $k \geq 0$ such that: $n = \theta \cdot 2^k$, θ odd

Modelling the Collatz Problem from a Jacobsthal Viewpoint

This corresponds to the usual extraction of the highest power of two dividing n (sometimes written $k = v_2(n)$). This observation suggests organising $\mathbb{N}$ into “binary columns” indexed by odd numbers. For each odd parameter θ , we consider the infinite sequence $\theta, 2\theta, 4\theta, 8\theta, \dots, \theta \cdot 2^k$, and denote its set of values by: $S(\theta) = \{\theta \cdot 2^k : k \geq 0\}$

Then the sets $S(\theta)$ form a partition of $\mathbb{N}$:

$$\mathbb{N} = \bigcup_{\theta \geq 1, \theta \text{ odd}} S(\theta), \quad S(\theta) \cap S(\delta) = \emptyset, \theta \neq \delta$$

In other words, every positive integer appears in exactly one such column.

The classical Jacobsthal numbers admit the closed form:

$$J_n = \frac{2^n - (-1)^n}{3} \quad (4)$$

In this article, we later introduce the quantities $m_{\theta,k}$ as natural generalisations of the Jacobsthal numbers, obtained by inserting a parameter θ . This is the origin of the term Jacobsthal-based used in this paper. Fix an odd integer θ , and consider the binary column associated with it. For each exponent $k \geq 0$, we introduce:

$$m_{\theta,k} = \frac{\theta \cdot 2^k - 1}{3} \quad (5)$$

It is not always an integer. However, whenever $m_{\theta,k} \in \mathbb{N}$, we obtain the identity:

$$3 m_{\theta,k} + 1 = \theta \cdot 2^k, \quad (6)$$

so $\theta \cdot 2^k$ is produced by a genuine Collatz operation $n \mapsto 3n+1$ applied to the odd integer $m_{\theta,k}$, followed by divisions by 2 inside the binary column $S(\theta)$. From a modular point of view, the integrality condition is immediate:

$$m_{\theta,k} \in \mathbb{N} \Leftrightarrow \theta \cdot 2^k \equiv 1 \pmod{3}. \quad (7)$$

Since $2 \equiv -1 \pmod{3}$, we have $2^k \equiv (-1)^k \pmod{3}$, hence:

$$\theta \cdot 2^k \equiv 1 \pmod{3} \Leftrightarrow \theta \equiv (-1)^k \pmod{3} \quad (8)$$

Therefore, if $\theta \equiv 0 \pmod{3}$, no node can occur. Otherwise, node positions are periodic with period 2 along each column: they occur for even k when $\theta \equiv 1 \pmod{3}$, and for odd k when $\theta \equiv 2 \pmod{3}$.

Remark : Eq. (5) is introduced directly from the inverse Collatz identity $3q + 1 = \theta \cdot 2^k$, so that $q = (\theta \cdot 2^k - 1)/3$, and we simply set $m_{\theta,k} := q$ whenever this quantity is an integer. The connection with Jacobsthal numbers is interpretative: for $\theta = 1$, admissible values of k yield the classical Jacobsthal closed form (up to the usual parity constraint), which motivates the term Jacobsthal-based.

The quantities $m_{\theta,k}$ introduced in the previous section single out special positions along each binary column. Whenever $m_{\theta,k}$ is an integer, it yields an explicit identity that connects two different columns. More precisely, if $m_{\theta,k} \in \mathbb{N}$, then $3 m_{\theta,k} + 1 = \theta \cdot 2^k$. Thus $\theta \cdot 2^k$ arises from a genuine Collatz operation followed by divisions by 2. We call a node any value of the form $\theta \cdot 2^k$ such that $m_{\theta,k}$ is an integer. Equivalently, a node is a value $\theta \cdot 2^k$ that can be written as $3q + 1$ with $q \in \mathbb{N}$:

$$\theta \cdot 2^k = 3q + 1 \quad (q = m_{\theta,k}). \quad (9)$$

Such values are special because they create an explicit link between two binary columns: the node lies on $S(\theta)$, while the integer q lies on the column determined by the odd part of q .

As k increases, node positions occur periodically along each column whenever $\theta \not\equiv 0 \pmod{3}$ (and no node occurs if $\theta \equiv 0 \pmod{3}$). Each time a node is encountered, it produces an integer q , hence a new odd parameter θ' (the odd part of q), and therefore a new column. Repeating this process generates a branching structure organised around powers of two.

This branching phenomenon is the core of the Jacobsthal-based construction. Instead of viewing the Collatz problem as a collection of isolated trajectories, we obtain a global network of arithmetic connections

organised by the binary decomposition. In the next section, we show how these connections can be displayed as a tree-like structure.

Results and Discussion

We can now describe the global branching structure suggested by the nodes introduced above. The guiding idea is to organise integers not by following the Collatz iteration forward, but by building, step by step, a network of explicit identities of the form $\theta \cdot 2^k = 3q + 1$, where θ is odd and $k \geq 0$. Each such identity links the binary column $S(\theta)$ to the binary column containing q .

A convenient way to present this construction is to start from the root column $\theta = 1$. Along the values $1 \cdot 2^k$ one looks for exponents k such that $m_{1,k} = (2^k - 1)/3 \in \mathbb{N}$. Whenever such an integer exists, it defines an arithmetic predecessor $q = m_{1,k}$ and therefore a new odd parameter θ' , namely the odd part of q .

The same procedure is then repeated for every newly created column. Each column $S(\theta)$ is scanned for its node positions (those k such that $m_{\theta,k}$ is an integer). Each node produces an integer $q = m_{\theta,k}$, whose odd part θ' determines a new column. Iterating this construction generates a branching structure organised by increasing powers of two.

A concrete example ($\theta = 5$)

We illustrate the construction on the binary column $S(5) = \{5 \cdot 2^k : k \geq 0\}$. We scan successive values of k and test whether $m_{5,k} = (5 \cdot 2^k - 1)/3$ is an integer.

For $k = 0$ we have $5 \cdot 2^0 = 5$, hence $m_{5,0} = (5 - 1)/3 = 4/3 \notin \mathbb{N}$, so no node is created at this level.

A first node occurs at $k = 1$, since $5 \cdot 2^1 = 10$ and $m_{5,1} = (10 - 1)/3 = 3 \in \mathbb{N}$. Thus 10 is a node and it yields the integer $q = 3$ through the identity $10 = 3 \cdot 3 + 1$.

Since $3 = 3 \cdot 2^0$, the odd part is $\theta' = 3$, meaning that the column $S(3)$ is attached to the column $S(5)$ at this node.

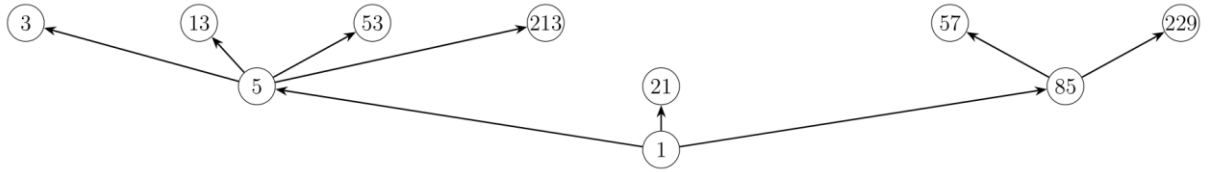


Fig. 1. A small spanning tree extracted from the Jacobsthal-based directed graph

In the next section, we explain how reading this tree in the reverse direction naturally recovers forward Collatz iterations.

So far, we have described a constructive process that builds a branching structure by scanning binary columns and creating new columns whenever an identity of the form $\theta \cdot 2^k = 3q + 1$ is satisfied. This construction proceeds “upward”, toward increasing powers of two. We now explain why this structure is naturally related to the usual Collatz iteration.

Suppose that a node occurs at the value $\theta \cdot 2^k$, meaning that there exists an integer q such that $\theta \cdot 2^k = 3q + 1$. Rewriting this identity gives $q = (\theta \cdot 2^k - 1)/3$. Thus, whenever $\theta \cdot 2^k \equiv 1 \pmod{3}$, the node $\theta \cdot 2^k$ determines a unique integer predecessor q .

To connect this with the Collatz map, write $q = \theta' \cdot 2^t$ with θ' odd, where $t = v_2(q)$ is the exponent of the highest power of 2 dividing q . Substituting into $3q + 1 = \theta \cdot 2^k$, we obtain $3\theta' \cdot 2^t + 1 = \theta \cdot 2^k$. In particular, when $t = 0$ (i.e., when q is odd), we have $3\theta' + 1 = \theta \cdot 2^k$ and:

$$T(\theta') = \frac{(3\theta' + 1)}{2^{v_2(3\theta' + 1)}} = \frac{(3\theta' + 1)}{2^t} = q \quad (10)$$

and multiplying by 2^k yields the node value $\theta \cdot 2^k = 3q + 1$. In other words, each edge of the tree encodes a genuine Collatz transition, but read in the reverse direction. Following a branch downward in the tree therefore reconstructs a forward Collatz trajectory. The accelerated iteration on odd integers $n \mapsto T(n)$ is obtained by descending along the edges, while the constructive process described in the previous sections builds the same relations in the opposite direction. This viewpoint shifts the perspective on the Collatz problem. Rather than studying isolated forward orbits, we obtain a global structure that simultaneously encodes many Collatz transitions. Classical trajectories appear as particular paths inside this structure.

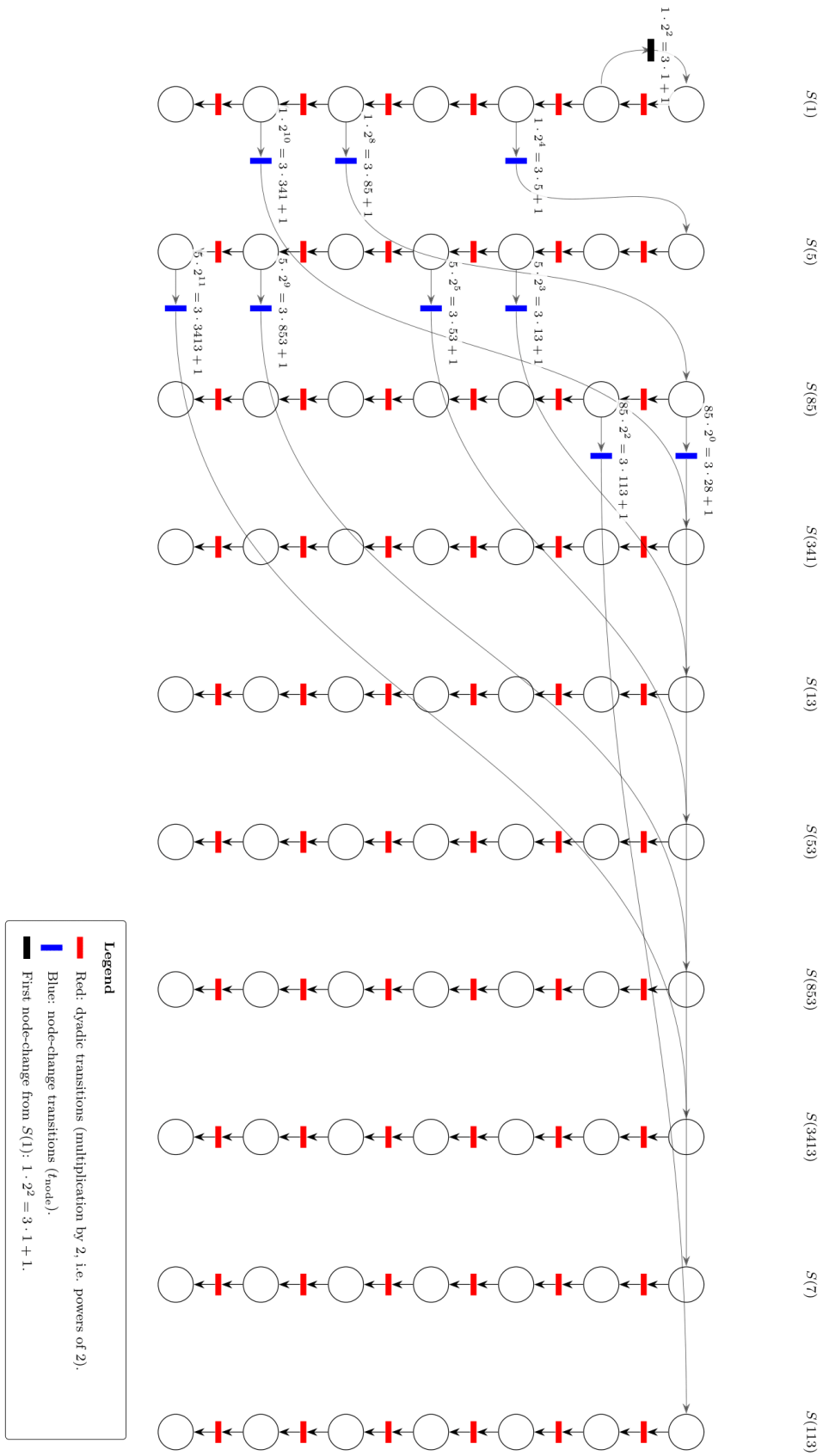


Fig. 2. Petri-net representation of the Collatz inverse structure, organized using a Jacobsthal-type construction

Example 1.

Consider the identity $10 = 3 \cdot 3 + 1$. This means that 10 is a node on the column $S(5)$, since $10 = 5 \cdot 2^1$. Reading this identity in reverse, we recover the predecessor $3 = (10 - 1)/3$, hence the Collatz step $3 \rightarrow 10$. Applying the usual rule to the odd integer 3, we obtain $3 \rightarrow 3 \cdot 3 + 1 = 10 \rightarrow 5$, which illustrates that after the $3n+1$ operation, one performs as many divisions by 2 as possible.

Example 2.

Consider the identity $16 = 3 \cdot 5 + 1$. Here $16 = 1 \cdot 2^4$ is a node on the column $S(1)$. Reading this identity in reverse yields $5 \rightarrow 16$. Continuing forward, we obtain the classical Collatz steps $5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$.

These examples show how usual Collatz trajectories appear as downward paths in the Jacobsthal tree. The constructive process described in the previous sections builds these transitions in advance, and forward iterations simply follow existing branches.

The Jacobsthal-based viewpoint modifies the way the problem is organised and interpreted. First, the present construction introduces a global structure in which many Collatz relations are encoded simultaneously. The binary decomposition $n = \theta \cdot 2^k$ provides a vertical organisation, while the identities $\theta \cdot 2^k = 3q + 1$ create horizontal connections between different columns.

Second, this global organisation makes it possible to visualise how different trajectories are related, how they merge, and how they are distributed among the binary columns. In this sense, the complexity is shifted from isolated forward iterates to the geometry of an explicit branching structure.

Why this is interesting : The main interest of this approach is conceptual rather than computational. It may serve as a basis for further investigations, for instance by studying the geometry of the tree, the distribution of nodes, or the way branches interact. Even without leading to a proof, this structural perspective provides new intuition and suggests alternative ways of thinking about the Collatz problem.

In practice, the Jacobsthal-based viewpoint can be formalized as a constructive algorithm, naturally represented as a Petri net, with each place corresponding to a dyadic level $\theta \cdot 2^k$. On the Figure 1 presented below, red transitions encode dyadic steps, while blue transitions encode node changes triggered by the condition $\theta 2^k \equiv 1 \pmod{3}$, that is, when $m_{\theta,k} = (\theta 2^k - 1)/3 \in \mathbb{N}$.

When such a transition fires, it generates a new value q satisfying $\theta 2^k = 3q + 1$, and spawns a new dyadic column associated with the odd part of q .

Conclusions

In this paper, we have presented a pedagogical and structural introduction to a Jacobsthal-based view of the Collatz problem in the classical case $\kappa = 3$. Rather than following individual forward iterates, we built a branching structure based on explicit arithmetic identities of the form $\theta \cdot 2^k = 3q + 1$.

This construction organises many inverse Collatz relations simultaneously and shows how usual trajectories can be recovered by a reverse reading of the resulting tree.

Although we restricted our exposition to the case $\kappa = 3$, the same philosophy naturally extends to more general odd multipliers. In particular, one may consider transformations of the form:

$$n \mapsto \frac{\kappa n + 1}{2^{v_2(\kappa n + 1)}}, \quad (11)$$

where κ is an odd integer and v_2 denotes the 2-adic valuation. Such extensions, developed in the work of P. Kosobutskyy, suggest that the constructive viewpoint presented here belongs to a broader family of arithmetic dynamical systems.

We hope that this elementary presentation will help make these ideas more accessible and will encourage further explorations of structural and geometric approaches to Collatz-type problems.

References

[1] Weisstein E. W. Collatz Problem. MathWorld—A Wolfram Web Resource. URL: <https://mathworld.wolfram.com/CollatzProblem.html>

Modelling the Collatz Problem from a Jacobsthal Viewpoint

- [2] Terras R. A stopping time problem on the positive integers. *Acta Arithmetica*, vol. 30, pp. 241–252, 1976. DOI: <https://doi.org/10.4064/aa-30-3-241-252>
- [3] Lagarias J. C. The $3x + 1$ problem and its generalizations. *The American Mathematical Monthly*, vol. 92, no. 1, pp. 3–23, 1985. DOI: <https://doi.org/10.1080/00029890.1985.11971528>
- [4] Lagarias J. C. The $3x + 1$ problem: An annotated bibliography (1963–1999). arXiv:math/0309224, Jan. 2011.
- [5] Matthews K. Generalized $3x + 1$ mappings: Markov chains and ergodic theory. *The Ultimate Challenge: The $3x + 1$ Problem*, J. C. Lagarias, Ed. Providence, RI, USA: American Mathematical Society, 2010, pp. 70–103.
- [6] Leigh G. A Markov process underlying the generalized Syracuse algorithm. *Acta Arithmetica*, vol. 46, pp. 125–143, 1986. DOI: <https://doi.org/10.4064/aa-46-2-125-143>
- [7] Chamberland M. A continuous extension of the $3x + 1$ problem to the real line. *Dynamical Systems*, vol. 2, no. 4, pp. 495–509, 1996.
- [8] Kosobutskyy P. The Collatz problem as a reverse problem on a graph tree formed from $Q \cdot 2^n$ ($Q = 1, 3, 5, 7, \dots$) Jacobsthal-type numbers. arXiv:2306.14635, 2023.
- [9] Kosobutskyy P., Rebot D. Collatz conjecture $3n \pm 1$ as a Newton binomial problem. *Computer Design Systems*, vol. 5, no. 1, pp. 137–145, 2023. DOI: <https://doi.org/10.23939/cds2023.01.137>
- [10] Kosobutskyy P. The Jacobsthal–Collatz–Terras model of conjecture: the natural numbers in $kq + 1$ problems. *Journal of Applied Mathematics*, vol. 3, p. 1767, 2025. DOI: <https://doi.org/10.59400/jam1767>
- [11] Mailland D., Grobelna I. A novel approach to the Collatz conjecture with Petri nets. *Information*, vol. 16, no. 7, p. 745, 2025. DOI: <https://doi.org/10.3390/info16090745>
- [12] Lagarias J. C. (Ed.). *The Ultimate Challenge: The $3x + 1$ Problem*. Providence, RI, USA: American Mathematical Society, 2010.

Conflict of interest

The authors declare that they have no conflict of interest.

Девід Мейланд¹, Петро Кособуцький²

¹ Незалежний дослідник з досвідом системного інженера з надійності Thales Alenia Space, Франція,
E-mail: david.mailland@protonmail.com, ORCID 0009-0002-8480-1325

² Кафедра систем автоматизованого проєктування, Національний університет “Львівська політехніка”,
вул. С. Бандери, 12, Львів, Україна, E-mail: petkosob@gmail.com, ORCID 0000-0003-4319-7395

МОДЕЛЮВАННЯ ЗАДАЧІ КОЛАТЦА З ТОЧКИ ЗОРУ ЯКОБСТАЛЯ

Отримано: Лютий 17, 2026 / Прийнято: Квітень 01, 2026 / Опубліковано: Квітень 30, 2026

© Мейланд Д., Кособуцький П., 2026*

Анотація. Ця публікація не має на меті розв’язати гіпотезу Коллатца, а переформулювати її в межах нової аналітичної структури, заснованої на послідовностях Якобсталя. Це переформулювання розкриває структурні закономірності та симетрії, які залишаються прихованими в традиційному підході, пропонуючи нову математичну перспективу, з якої можна далі досліджувати проблеми типу Коллатца. Цей підхід може бути відтворений в аналізі системної інженерії, де альтернативні представлення часто є цінними для виявлення прихованої структури та покращення розуміння. Окрім математичного значення, подана структура також може запропонувати концептуальні та методологічні переваги для галузі інформаційних технологій (ІТ). Багато проблем в інформатиці пов’язані з аналізом ітераційних процесів, рекурсивних алгоритмів та дискретних переходів станів. Саме перетворення Коллатца можна розглядати як простий обчислювальний процес, що регулюється умовними правилами та повторюваною ітерацією, що робить його цікавою моделлю для вивчення алгоритмічної поведінки. Представлення на основі послідовностей широко використовуються в таких галузях, як стиснення даних, криптографія та генерація псевдовипадкових чисел. Ідентифікація структурованих рекурентних співвідношень та шаблонів парності в цілочисельних послідовностях може дати уявлення про розробку ефективних алгоритмів для генерації детермінованих, але складних числових послідовностей. Такі послідовності часто використовуються в криптографічних системах, імітаційних моделях та розподілених обчислювальних програмах.

Ключові слова: модельний аналіз, дискретні динамічні системи, гіпотеза Коллатца, послідовності Якобсталя, формулювання оберненої задачі, структурні закономірності, модульна арифметика, 2-адична оцінка, відтворювані обчислення.

